Classification of Cancer outcome using Genetic and Clinical data

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Exploratory Data Analysis

Introduction

This project investigates the effect of genetic and clinical variables on the survival outcome of breast cancer patients. The data contains records of more than 1000 breast cancer patients from several research institutions. Clinical data contains patient-related and tumor-related information. Additionally mRNA gene expression data is available for each patient. The gene expression data has been processed to include only the top 5000 most variable genes on the transformed scale (log2(counts + 1)).

We consider that the main outcome variable of interest is **vital_status**, which is defined in clinical data.

Load data

```
load("mrr_bio.Rdata")
```

Genes data

```
dim(GeneX)
```

```
## [1] 1231 5000
```

```
str(GeneX)
```

```
##  num [1:1231, 1:5000] 0 0 10.92 3.17 5 ...
##  - attr(*, "dimnames")=List of 2
##   ..$ : chr [1:1231] "EW-A2FS-01A-11R-A17B" "OL-A6VR-01A-32R-A33J" "E9-A226-01A-21R-A157" "A8-A08H-0
##   ..$ : chr [1:5000] "CLEC3A" "SCGB2A2" "CPB1" "GSTM1" ...
```

```
colnames(GeneX)[1:5]
```

```
## [1] "CLEC3A"  "SCGB2A2" "CPB1"    "GSTM1"   "TFF1"
```

```
rownames(GeneX)[1:5]
```

```
## [1] "EW-A2FS-01A-11R-A17B" "OL-A6VR-01A-32R-A33J" "E9-A226-01A-21R-A157"
## [4] "A8-A08H-01A-21R-A00Z" "D8-A27H-01A-11R-A16F"
```

```
head(GeneX[, 1:5])
```

```
##                         CLEC3A   SCGB2A2      CPB1    GSTM1      TFF1
## EW-A2FS-01A-11R-A17B  0.000000 16.259357 18.018521  0.00000 13.656872
```

```
## OL-A6VR-01A-32R-A33J  0.000000 16.522045  7.189825 12.77992 12.802920
## E9-A226-01A-21R-A157 10.915132  5.285402  5.930737 10.22279 10.195987
## A8-A08H-01A-21R-A00Z  3.169925  9.952741  7.734710  0.00000  1.584963
## D8-A27H-01A-11R-A16F  5.000000  7.643856  7.467606  1.00000  0.000000
## D8-A3Z6-01A-11R-A239 15.954469 13.971723 13.489346  1.00000 16.881711
```

`GeneX` is a table with 1231 rows and 5000 columns:

- The rows correspond to the patients with each row name being the patient identifier such as `EW-A2FS-01A-11R-A17B`.
- The columns correspond to the genes with each column name being a gene such as `CLEC3A`, which are the 5000 most variable genes from one person to another.
- Each cell corresponds to the expression value (a float) of a certain gene (column) for a certain patient (row) -> continuous variables

Clinical data

```r
load("mrr_bio.Rdata")
dim(clinical_data)
```

```
## Loading required namespace: S4Vectors
```

```
## NULL
```

```r
head(clinical_data[, 1:5])
```

```
## DataFrame with 6 rows and 5 columns
##                      initial_weight tissue_type  laterality
##                           <numeric> <character> <character>
## EW-A2FS-01A-11R-A17B            110       Tumor        Left
## OL-A6VR-01A-32R-A33J            380       Tumor       Right
## E9-A226-01A-21R-A157            510       Tumor        Left
## A8-A08H-01A-21R-A00Z            250       Tumor        Left
## D8-A27H-01A-11R-A16F            170       Tumor       Right
## D8-A3Z6-01A-11R-A239             60       Tumor        Left
##                      tissue_or_organ_of_origin age_at_diagnosis
##                                    <character>        <integer>
## EW-A2FS-01A-11R-A17B              Breast, NOS            15302
## OL-A6VR-01A-32R-A33J              Breast, NOS            17702
## E9-A226-01A-21R-A157              Breast, NOS            16511
## A8-A08H-01A-21R-A00Z              Breast, NOS            24137
## D8-A27H-01A-11R-A16F              Breast, NOS            26588
## D8-A3Z6-01A-11R-A239              Breast, NOS            20629
```

```r
colnames(clinical_data)
```

```
##  [1] "initial_weight"              "tissue_type"
##  [3] "laterality"                  "tissue_or_organ_of_origin"
##  [5] "age_at_diagnosis"            "primary_diagnosis"
##  [7] "prior_treatment"             "diagnosis_is_primary_disease"
##  [9] "ajcc_pathologic_t"           "morphology"
## [11] "classification_of_tumor"     "sites_of_involvement"
## [13] "days_to_last_follow_up"      "follow_ups_disease_response"
## [15] "race"                        "gender"
## [17] "ethnicity"                   "vital_status"
## [19] "age_at_index"                "days_to_birth"
```

```
## [21] "age_is_obfuscated"             "bcr_patient_barcode"
## [23] "primary_site"                  "disease_type"
```

`clinical_data` is a table with 1231 rows and 24 columns:

- The rows correspond to the names of the patients, similar to `GeneX`.
- The columns are the names of the clinical data, such as `initial_weight`, `sites_of_involvement`, and `vital_status`.
- Each cell corresponds to the value of the clinical data for each patient, which can be either an integer or a character string.

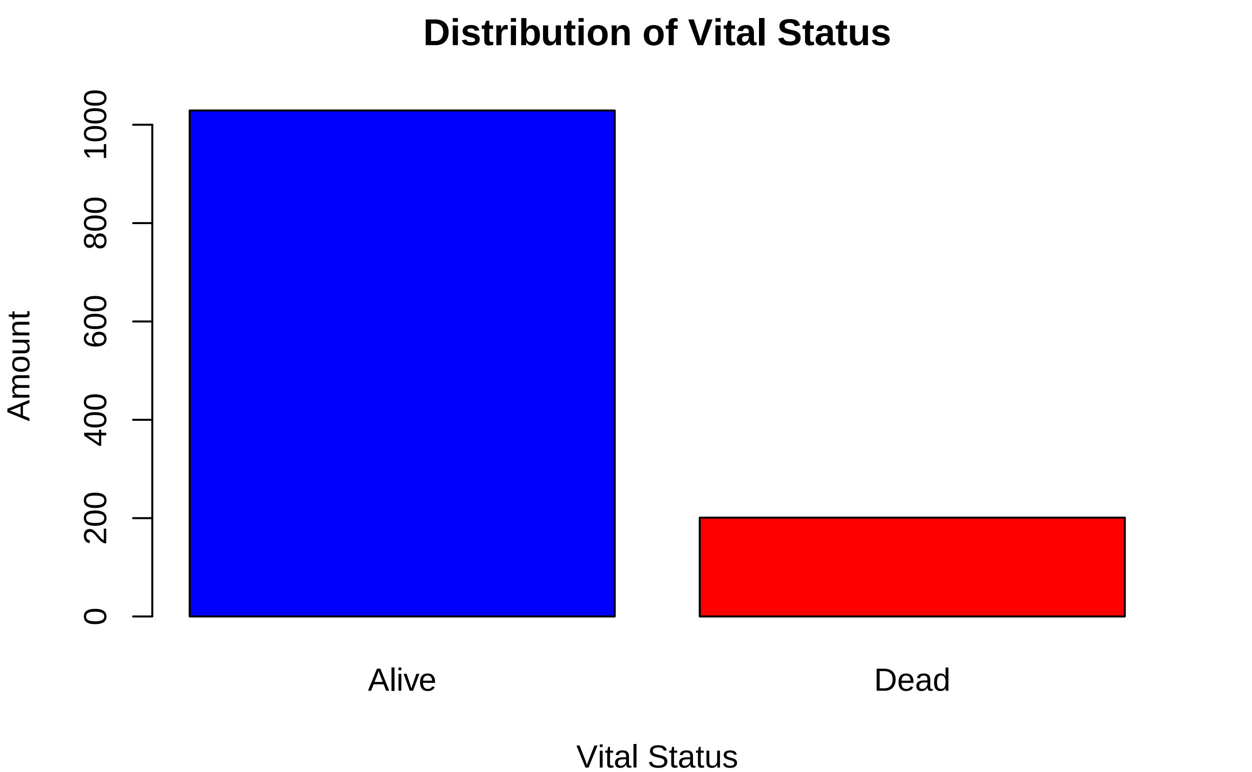
Outcome variable

The outcome variable is the vital status of the patient (dead or alive).

```r
y <- clinical_data$vital_status
table(y)
```

```
## y
## Alive  Dead
##  1029   201
```

```r
barplot(table(y),
        main = "Distribution of Vital Status",
        xlab = "Vital Status",
        ylab = "Amount",
        col = c("blue", "red"))
```

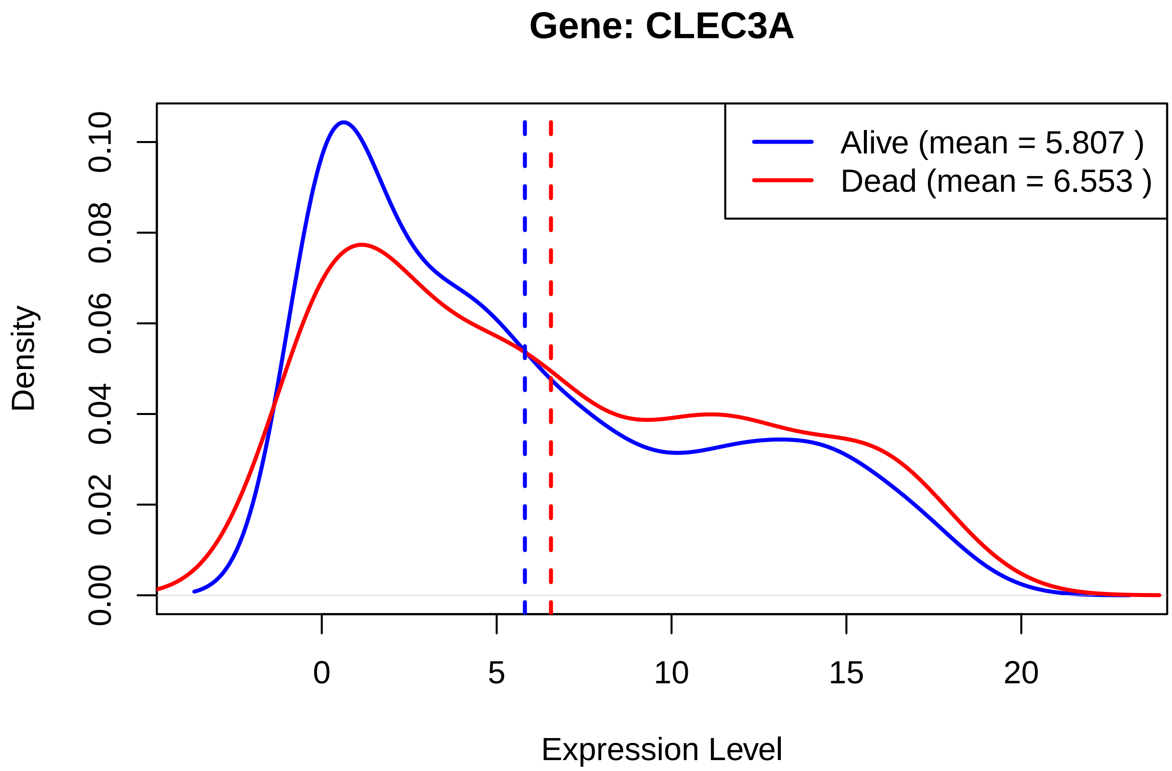


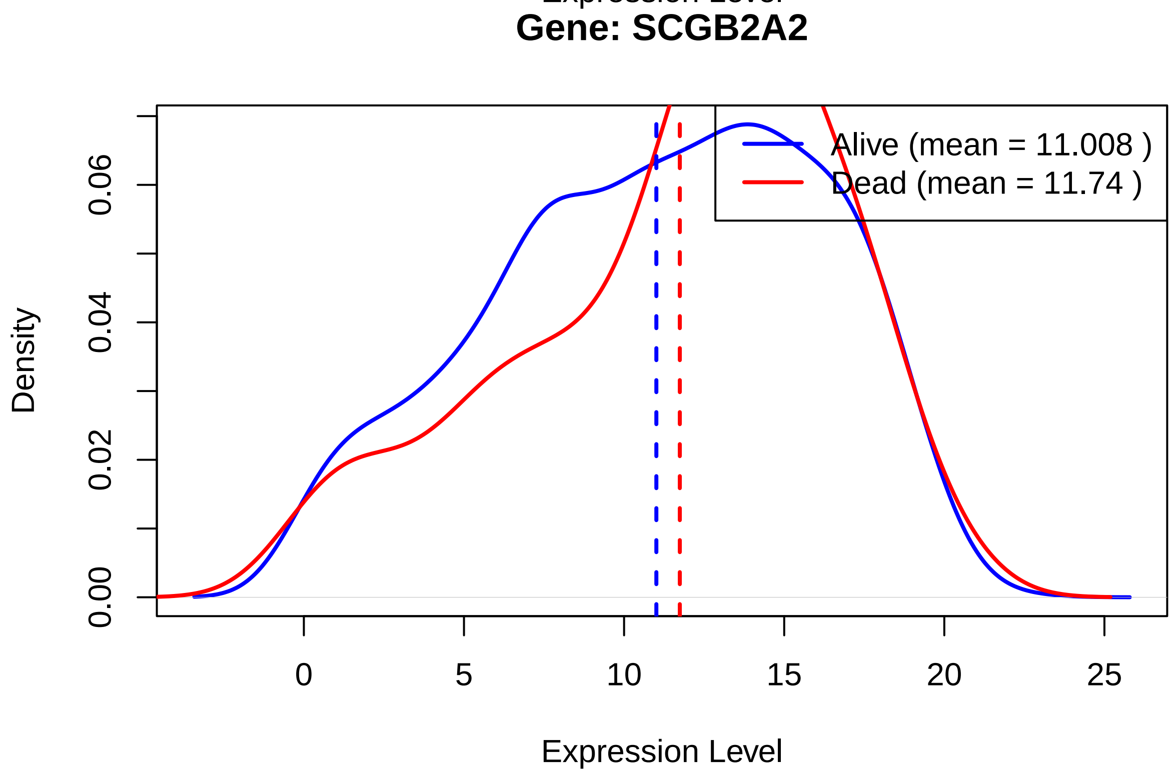
Exploratory data analysis

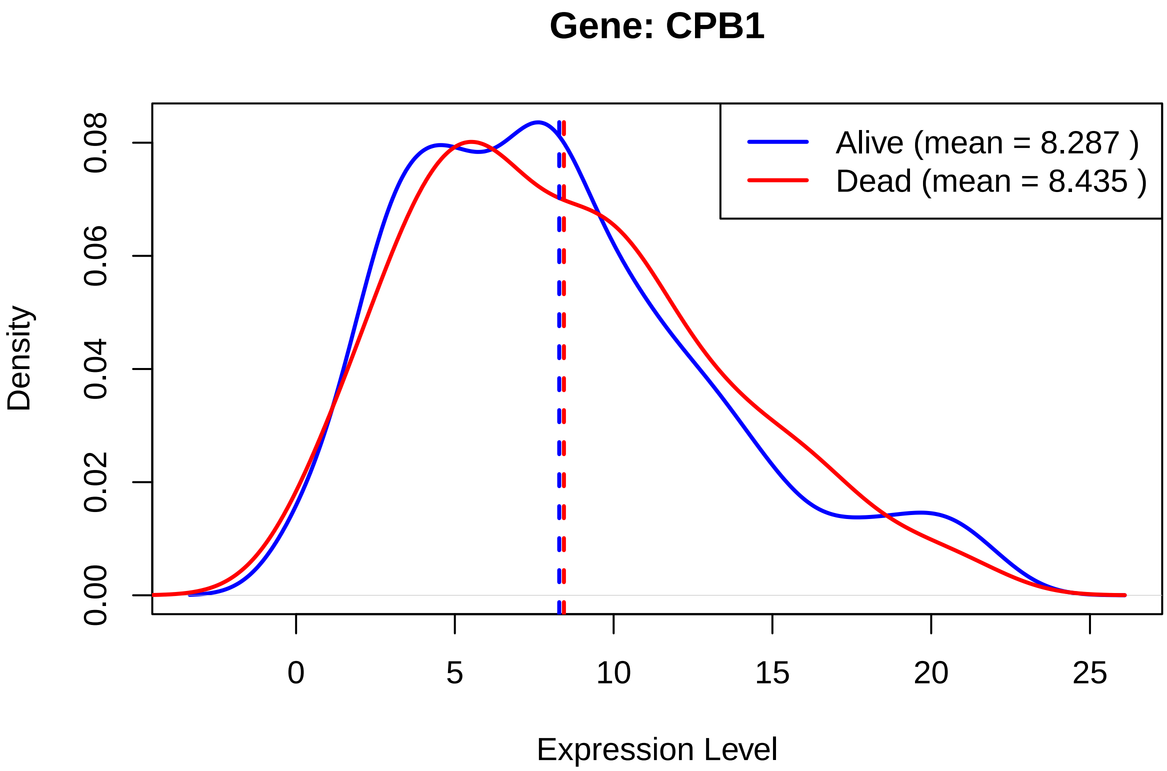
Genes

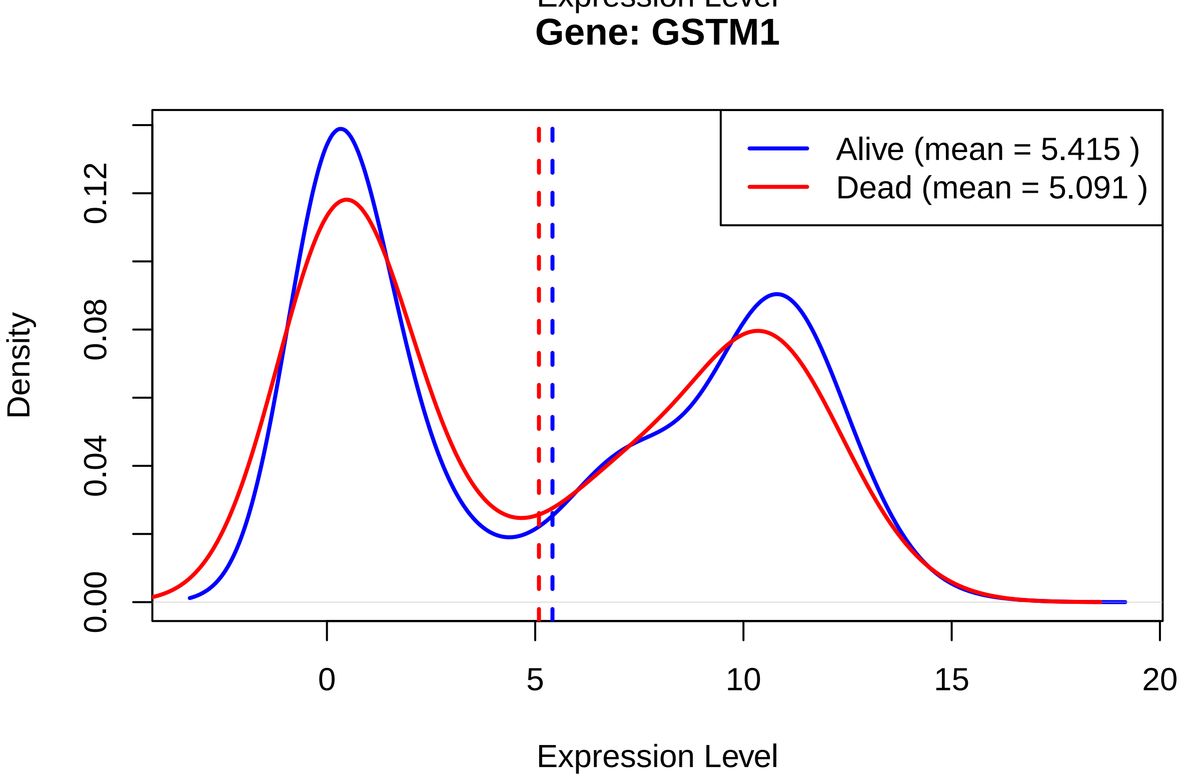
For the genes table, let's first plot for each gene the gaussian distribution of proportion of dead minus proportion of alive and then take the average of this graph.

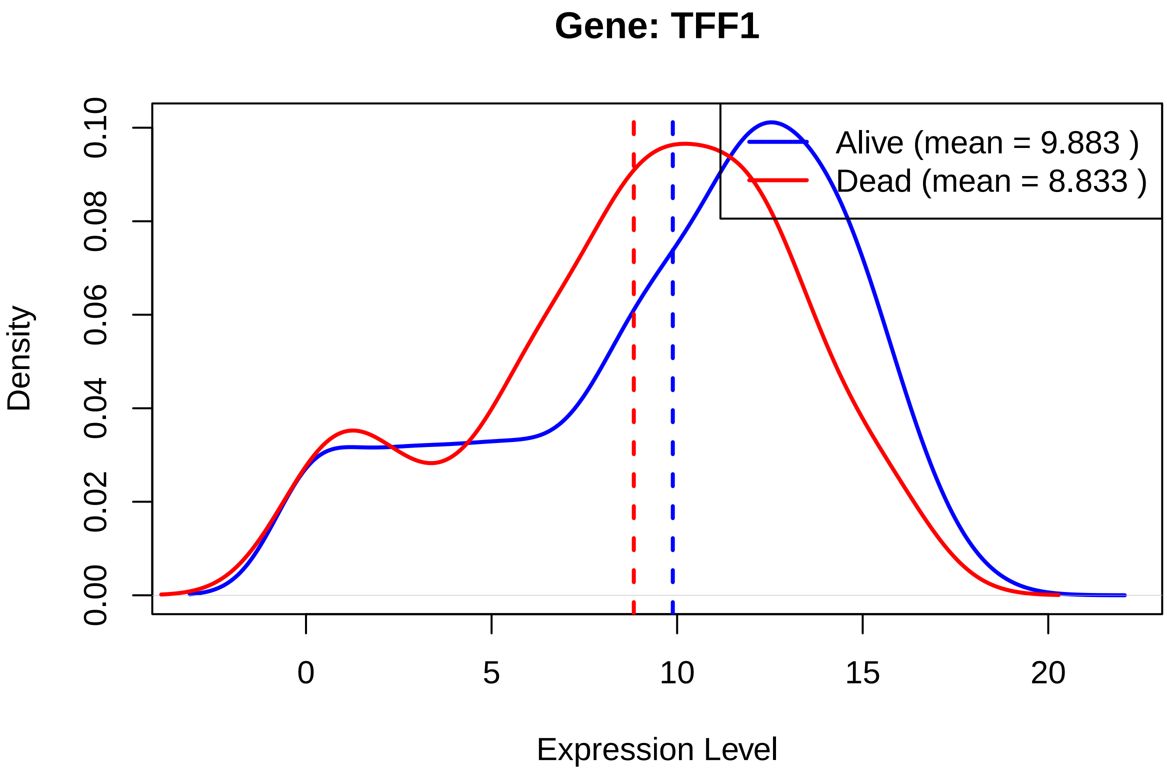
```r
# Calculate mean expression for each gene by vital status
for (col in colnames(GeneX)[1:5]) {  # First 5 genes for demonstration
  gene_data <- GeneX[, col]
  status <- clinical_data$vital_status

  # Split data by status
  alive_expr <- gene_data[status == "Alive"]
  dead_expr <- gene_data[status == "Dead"]

  # Calculate means
  mean_alive <- mean(alive_expr, na.rm = TRUE)
  mean_dead <- mean(dead_expr, na.rm = TRUE)

  # Create density plot
  plot(density(alive_expr, na.rm = TRUE),
       col = "blue",
       lwd = 2,
       main = paste("Gene:", col),
       xlab = "Expression Level",
       ylab = "Density")
  lines(density(dead_expr, na.rm = TRUE),
        col = "red",
        lwd = 2)

  # Add mean lines
  abline(v = mean_alive, col = "blue", lty = 2, lwd = 2)
  abline(v = mean_dead, col = "red", lty = 2, lwd = 2)

  # Add legend
  legend("topright",
         legend = c(paste("Alive (mean =", round(mean_alive, 3), ")"),
                    paste("Dead (mean =", round(mean_dead, 3), ")")),
         col = c("blue", "red"),
         lty = 1,
         lwd = 2)
}
```


Gene: CLEC3A
Alive (mean = 5.807)
Dead (mean = 6.553)
Density
0.00
0.02
0.04
0.06
0.08
0.10
0
5
10
15
20
Expression Level


Gene: SCGB2A2
Alive (mean = 11.008)
Dead (mean = 11.74)
Density
0.00
0.02
0.04
0.06
0
5
10
15
20
25
Expression Level


Gene: CPB1
Alive (mean = 8.287)
Dead (mean = 8.435)
Density
0.00
0.02
0.04
0.06
0.08
0
5
10
15
20
25
Expression Level


Gene: GSTM1
Alive (mean = 5.415)
Dead (mean = 5.091)
Density
0.00
0.04
0.08
0.12
0
5
10
15
20
Expression Level



Thus we can display the most important genes by calculating the relative mean difference between the two groups (alive and dead) for each gene and then sorting them.

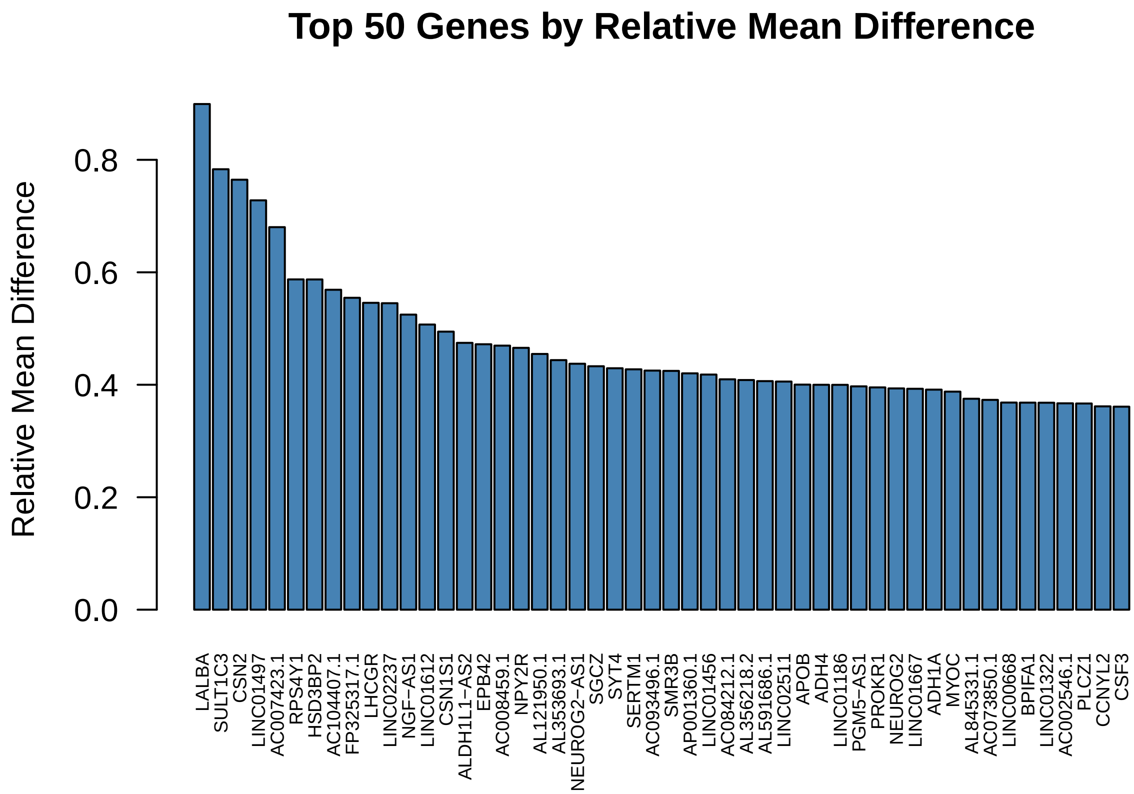
```r
gene_diff <- data.frame(
  Gene = colnames(GeneX),
  MeanDiff = numeric(ncol(GeneX))
)

for (i in seq_along(colnames(GeneX))) {
  col <- colnames(GeneX)[i]
  gene_data <- GeneX[, col]
  status <- factor(clinical_data$vital_status)

  plot_data <- data.frame(Expression = gene_data, Status = status)
  means <- aggregate(Expression ~ Status, data = plot_data, mean)

  # Calculate relative mean difference
  mean_alive <- means$Expression[means$Status == "Alive"]
  mean_dead <- means$Expression[means$Status == "Dead"]

  # Relative difference: (Dead - Alive) / Alive
  # Handle division by zero
  if (mean_alive != 0) {
    gene_diff$MeanDiff[i] <- abs((mean_dead - mean_alive) / mean_alive)
  } else {
    gene_diff$MeanDiff[i] <- NA
  }
}

# Sort by relative mean difference
```

```r
gene_diff <- gene_diff[order(gene_diff$MeanDiff, decreasing = TRUE), ]

# Display top 50 genes
top_50 <- head(gene_diff, 50)
print(top_50)
```

```
##             Gene  MeanDiff
## 900        LALBA 0.8992166
## 1506     SULT1C3 0.7832339
## 2589        CSN2 0.7646311
## 4892   LINC01497 0.7279608
## 3569  AC007423.1 0.6802403
## 2308      RPS4Y1 0.5873672
## 2128     HSD3BP2 0.5872528
## 737   AC104407.1 0.5689319
## 1657  FP325317.1 0.5546457
## 2995       LHCGR 0.5456845
## 3509   LINC02237 0.5450194
## 3699     NGF-AS1 0.5247317
## 4982   LINC01612 0.5070537
## 748       CSN1S1 0.4944819
## 2205 ALDH1L1-AS2 0.4744773
## 3488       EPB42 0.4720326
## 1133  AC008459.1 0.4695645
## 192        NPY2R 0.4656152
## 3270  AL121950.1 0.4547892
## 3963  AL353693.1 0.4437907
## 4992 NEUROG2-AS1 0.4373508
## 4097        SGCZ 0.4328951
## 1595        SYT4 0.4293564
## 3806      SERTM1 0.4274395
## 3696  AC093496.1 0.4251907
## 587        SMR3B 0.4246711
## 3893  AP001360.1 0.4203018
## 3438   LINC01456 0.4180961
## 4182  AC084212.1 0.4096885
## 1784  AL356218.2 0.4084072
## 2672  AL591686.1 0.4064550
## 2057   LINC02511 0.4056917
## 743         APOB 0.4002554
## 2201        ADH4 0.3999677
## 3164   LINC01186 0.3998710
## 3053    PGM5-AS1 0.3971454
## 2269      PROKR1 0.3952935
## 974      NEUROG2 0.3935187
## 1003   LINC01667 0.3928182
## 2475       ADH1A 0.3913356
## 525         MYOC 0.3877588
## 2289  AL845331.1 0.3750996
## 2805  AC073850.1 0.3730908
## 1092   LINC00668 0.3683282
## 3341      BPIFA1 0.3681901
## 4672   LINC01322 0.3680301
## 1430  AC002546.1 0.3669965
```

```
## 3273       PLCZ1 0.3666600
## 4006      CCNYL2 0.3616090
## 1772        CSF3 0.3610313
```

```r
# Optional: visualize the distribution
barplot(top_50$MeanDiff[1:50],
        names.arg = top_50$Gene[1:50],
        las = 2,
        cex.names = 0.6,
        main = "Top 50 Genes by Relative Mean Difference",
        ylab = "Relative Mean Difference",
        col = "steelblue")
```



Clinical Data

First modeling attempt with three variables from clinical data : `initial_weight`, `classification_of_tumor`, `morphology` to predict `vital_status`.

```r
x <- clinical_data[, c("initial_weight",
                       "classification_of_tumor",
                       "morphology")]
y <- clinical_data$vital_status
model_data <- data.frame(x, vital_status = factor(y))
model_data <- na.omit(model_data) # removes NA for now
model_data$vital_status <- droplevels(model_data$vital_status)
```

Model fitting:

```r
model <- glm(vital_status ~ ., data = model_data, family = binomial)
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

Prediction:

```r
pred <- predict(model, type = "response")
positive_class <- levels(model_data$vital_status)[2]
pred_class <- ifelse(pred > 0.5, positive_class, levels(model_data$vital_status)[1])
pred_class <- factor(pred_class, levels = levels(model_data$vital_status))
```

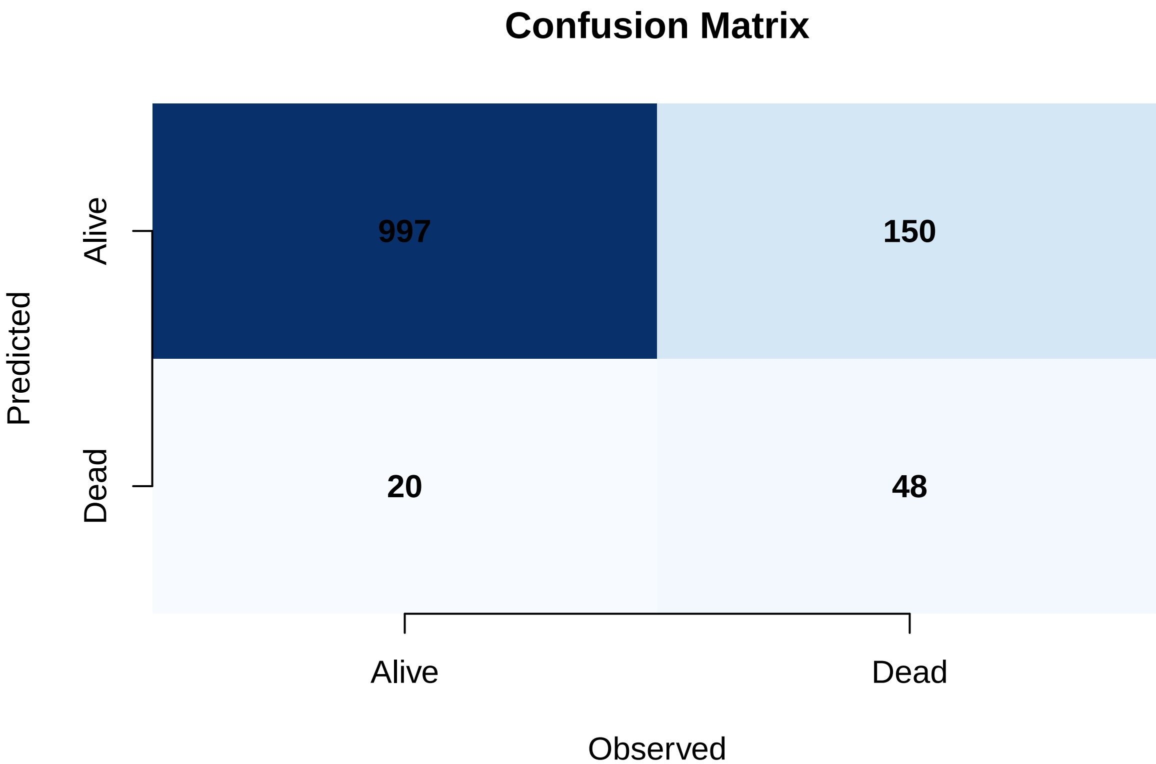
Confusion matrix:

```r
conf_mat <- table(Predicted = pred_class, Observed = model_data$vital_status)
conf_mat
```

```
##          Observed
## Predicted Alive Dead
##     Alive   997  150
##     Dead     20   48
```

Visualization of the confusion matrix.

```r
conf_mat_matrix <- as.matrix(conf_mat)

image(t(conf_mat_matrix[nrow(conf_mat_matrix):1, ]),
  axes = FALSE,
  col = colorRampPalette(c("#f7fbff", "#6baed6", "#08306b"))(64),
  xlab = "Observed",
  ylab = "Predicted",
  main = "Confusion Matrix")

axis(1, at = seq(0, 1, length.out = ncol(conf_mat_matrix)), labels = colnames(conf_mat_matrix))

axis(2, at = seq(0, 1, length.out = nrow(conf_mat_matrix)), labels = rev(rownames(conf_mat_matrix)))

text(rep(seq(0, 1, length.out = ncol(conf_mat_matrix)), each = nrow(conf_mat_matrix)),
     rep(seq(0, 1, length.out = nrow(conf_mat_matrix)), times = ncol(conf_mat_matrix)),
     labels = as.vector(conf_mat_matrix[rev(seq_len(nrow(conf_mat_matrix))), ]),
     col = "black", font = 2)
```


Confusion Matrix
Predicted
Alive
Dead
997
150
20
48
Alive
Dead
Observed